

MS-AIB Applied Project Report

The Vinyl Concierge

Final Project Business Report

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W. P. Carey School of Business Honor Code

As a W. P. Carey student, I am committed to honesty, integrity, professional courtesy, and responsible use of AI-supported work. This report synthesizes my prior course submissions, prototype development work, instructor feedback, and project evidence into a final business-facing product judgment document.

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Executive Summary

The Vinyl Concierge is an AI-powered recommendation and decision-support platform for independent record stores. It addresses two connected problems: customers often want warmer, more personal music discovery than generic streaming algorithms provide, while store owners must make inventory decisions with limited visibility into emerging customer interests. The product connects those two needs by turning customer recommendation activity into de-identified owner intelligence.

The current prototype lets customers sign in, enter favorite artists or genres, receive five vinyl recommendations from a limited demo catalog, and respond with saved-interest, thumbs-up, or thumbs-down signals. Those interactions can be saved in Firebase and converted into owner-facing demand signals. The Owner Intelligence Dashboard then reviews album, artist, and genre patterns as cautious sourcing signals rather than automated purchase instructions.

The project evolved from a prompt-centered recommendation idea into a fuller business workflow. Earlier versions emphasized atmosphere and record-store personality. Later work added a clearer product requirements document, Collection Insights, an Owner Intelligence Dashboard, Firebase Authentication, Firestore persistence, security rules, feedback collection, bias analysis, governance mapping, and an ROI/deployment judgment. Human and AI feedback also led to important product changes: simpler language, larger type, clearer match-confidence labels, removal of cart-like actions, and stronger prototype notices.

The main business conclusion is that The Vinyl Concierge is worth a limited pilot, not broad deployment. The prototype demonstrates a coherent customer-to-owner feedback loop, but it is not yet a production SaaS product. A commercial version would need live inventory integration, POS or ecommerce connections, purchase validation, stronger monitoring, and store-specific rollout support. The strongest value proposition is not the recommendation engine alone; it is the decision-support loop that connects customer discovery, de-identified interest signals, owner review, and future inventory planning.

Known limitations are material but manageable if the product remains transparent. Recommendation activity is not the same as purchase intent; the demo catalog is limited; the current interface assumes English-speaking users comfortable with record-store vocabulary; and owner dashboards could create automation bias if signals are overstated. Responsible deployment therefore requires clear labeling, role-based access, de-identified owner analytics, human review, and governance triggers for complaints, data drift, suspicious demand spikes, and inventory mismatches.

Product Overview

Core Problem

Independent record stores operate in a retail category where discovery, trust, and inventory judgment matter. Customers may know a mood, artist, or listening context but still need help finding records that feel right. Store owners, meanwhile, need to decide what to stock, restock, display, or special order before clear sales patterns appear. These decisions are often made with staff expertise, historical sales, and intuition. That expertise is valuable, but it can miss early signals of emerging demand or underserved genres.

The Vinyl Concierge frames recommendation activity as a business signal. Instead of treating AI recommendations as an isolated customer feature, the product asks whether customer discovery behavior can also help a store owner understand demand. The core product logic is that the same interaction that helps a customer discover records can, if handled responsibly, create de-identified signals for inventory planning.

User	Problem	Product Response
Customer	Wants music discovery that feels personal, explainable, and connected to a record-store experience.	Enters artists, genres, mood, or listening context and receives a five-record recommendation stack with plain-language explanations.
Store owner	Needs better visibility into emerging customer interests before making inventory decisions.	Reviews de-identified recommendation activity, saved-interest responses, and cautious sourcing suggestions in Owner Intelligence.
Business	Needs the recommendation experience to create durable value beyond a single customer session.	Connects customer discovery to owner decision support while keeping purchasing decisions under human review.

Value Proposition

For customers, the value is a warmer discovery experience: the system behaves less like a generic filter and more like a knowledgeable shop clerk who can explain why a record fits. For owners, the value is earlier visibility into what customers are asking for, saving, liking, or rejecting. For the business, the defensible opportunity is the feedback loop between customer recommendation behavior and store-specific decision-making.

This distinction matters. A recommendation interface alone is easy to copy. A store-specific intelligence loop, reinforced by inventory data, purchase behavior, customer feedback, and staff judgment, is harder to replicate and more valuable to independent stores.

Product Evolution and Design Trade-offs

The project evolved across the course from a customer-facing AI recommendation idea into a prototype that includes business intelligence, backend persistence, feedback integration, bias analysis, and deployment judgment. The most important evolution was a shift from output generation to decision support.

Stage	What Changed	Why It Mattered
Initial concept	Focused on a warm AI record-store clerk that recommends albums from customer preferences.	Established the customer experience and brand voice, but risked becoming only a themed recommendation engine.
PRD and early prototype	Defined measurable functionality and introduced the customer-to-owner feedback loop.	Made the product logic stronger by connecting recommendations to business value.
Owner Intelligence	Added a dashboard workflow for identifying inventory issues, validating signals, reviewing recommendations, and tracking outcomes.	Shifted the owner side from decorative analytics to a decision workflow.
Backend and governance	Added Firebase Authentication, Firestore catalog reads, private sessions, de-identified demand signals, and role-sensitive owner access.	Moved the product beyond prompt logic into persistence, privacy, and accountability.
Feedback and bias work	Integrated tester feedback, improved readability, clarified prototype limitations, corrected ranking issues, and documented bias risks.	Improved trust and usability while reducing overclaiming.
Final positioning	Added prototype notices and a commercial roadmap while keeping live inventory, checkout, and POS integration deferred.	Clarified what is built now versus what belongs in a production SaaS version.

Key Trade-offs

- Decision support over automation: The owner dashboard presents sourcing suggestions for review, but it does not automatically trigger buying decisions.

- Limited demo catalog over live inventory: The prototype uses a controlled catalog so recommendation logic can be tested safely, while live inventory remains future work.
- Record-store personality over generic SaaS tone: The interface keeps a warm Curate identity, but tester feedback led to simpler language where niche terminology created friction.
- Interest signals over purchase claims: Saved interest and thumbs responses are useful, but they are not treated as confirmed demand or sales forecasts.
- Temporary localization as learning, not final scope: The Week 6 Mexico-focused route was useful for bias and global-audience analysis, but it is not part of the final prototype experience.

Core Functionality and User Experience

The current GitHub Pages prototype includes two connected experiences: a customer recommendation flow and an owner intelligence flow. The customer side is the entry point, while the owner side demonstrates how de-identified recommendation activity can become business intelligence.

Customer Flow

- Step 1: The customer opens the prototype and sees a Prototype Notice explaining that recommendations use a demo catalog, not live store inventory.
- Step 2: The customer can sign in with Google and submit either favorite artists or genre choices, with mood and listening context available as optional refinements.
- Step 3: The recommendation engine returns five records with match labels and plain-language shelf notes.
- Step 4: The customer can save interest, like, or dislike individual recommendations.
- Step 5: If signed in, the session and response signals can be saved to Firestore and carried into Owner Intelligence.

Owner Flow

- Step 1: The owner opens the Owner Intelligence Dashboard from the app header or from a specific customer-session handoff.
- Step 2: The dashboard distinguishes live de-identified interest signals from synthetic inventory, revenue, and outcome examples.
- Step 3: The owner reviews album, artist, and genre patterns as directional sourcing evidence.
- Step 4: The owner considers warnings about sample size, signal quality, and whether interest is strong enough to justify action.
- Step 5: Any inventory decision remains a human decision and should be validated against actual sales, stock levels, budget, and staff knowledge.

The professor walkthrough path demonstrates this loop directly: a tester signs in, generates recommendations, marks interest or feedback on a record, and opens the same session in Owner Intelligence. This is the strongest current evidence that the prototype has moved beyond a static interface into a connected workflow.

System Perspective

The current prototype is a React and Vite web application with Firebase Authentication and Firestore. It can read a Firestore-backed album catalog, fall back to local catalog data if needed, save private customer sessions under each user's profile, and write de-identified owner-readable demand signals. Firestore security rules protect role changes, validate demand-signal fields, and keep private recommendation actions under the customer's user path.

Current prototype data flow:

- Step 1: Customer preference input
- Step 2: Recommendation scoring against demo catalog
- Step 3: Recommendation result display with match-confidence labels
- Step 4: Private session save under user profile
- Step 5: De-identified demand signal creation
- Step 6: Owner Intelligence aggregation and review

A production version would preserve this architecture but replace the demo catalog with live inventory and commerce data. The future system would connect inventory availability, checkout behavior, POS history, and customer actions so that owner recommendations are grounded in both interest and purchase evidence. That production layer is intentionally not claimed as complete in the final prototype.

Built in Prototype	Future Commercial Version
Demo catalog plus Firestore catalog reads	Live inventory synchronization
Signed-in customer sessions	Customer history across visits and purchases
De-identified recommendation interest signals	Validated demand combining interest, sales, and stock movement
Owner Intelligence review dashboard	Operational inventory, merchandising, and buying workflows
Prototype notices and governance rules	Monitoring, incident response, and store-level admin tools

Validation and Feedback Integration

The prototype was shaped by both internal AI review and external human feedback. Internally, AI-assisted reviews identified implementation and product risks: role escalation, misleading demand signals, recommendation overconfidence, and interface elements that looked more production-ready than they were. These findings led to stronger Firestore rules, clearer demand-signal validation, match-confidence labels, and revised interface language.

Externally, ten participants tested the prototype and provided feedback through a form connected to a response sheet. The participants included a graphic/UX designer, classmates, eLearning designers, music buyers, and friends. The most repeated feedback involved readability, confusing record-store terminology, unclear input requirements, uncertainty about nonfunctional cart/audio expectations, and the need to make recommendation confidence easier to understand.

Feedback Pattern	Action Taken
Text and spacing felt dense	Increased type size and improved spacing, including mobile spacing.
Users were unsure which inputs were required	Allowed recommendations from either artists or genres; mood and context remain optional.
Cart language implied ecommerce functionality	Replaced cart-style actions with prototype-only Save Interest and response controls.
Recommendation confidence was unclear	Added labels such as exact artist hit, adjacent fit, and limited-catalog suggestion.
Live Display Bins confused users	Removed that panel from the customer view and emphasized broader Suggested Exploration Areas.
Some terminology felt too niche	Simplified result language without removing the overall Curate personality.

This feedback process strengthened the final product argument. It showed that users liked the overall identity and recommendation concept, but trust depended on the product being honest about what it could do. The final prototype therefore prioritizes transparent signals over inflated claims.

Limitations, Bias, and Risk

Known Limitations

- The current catalog is limited and does not represent a complete store inventory or a universal music catalog.
- Recommendation activity is an interest signal, not confirmed purchase intent.

- The owner dashboard still includes synthetic inventory, revenue, and outcome examples for demonstration continuity.
- The product currently assumes English-language users who can describe taste through artists, genres, moods, or listening contexts.
- The prototype does not include live POS, ecommerce, checkout, purchase analytics, payment, or shipping integrations.

Bias and Transparency

The main bias risks are representation bias, assumption bias, algorithmic availability bias, language and cultural bias, and automation bias. These risks are not abstract. Earlier testing showed that broad catalog tags could interfere with more specific country or classic-rock requests until the scoring logic was corrected. Similarly, Week 6 localization work showed that language translation alone is not enough; product tone, genre categories, trust cues, privacy expectations, and local buying habits all affect whether a recommendation system fits a market.

For the final version, the Mexico-focused route is treated as temporary course evidence rather than part of the final prototype. Its lesson is still important: responsible deployment requires awareness of blind spots and local context. A production rollout outside the original U.S. independent-record-store context would need localized catalogs, genre taxonomies, privacy review, and culturally appropriate customer language.

Risk Controls

- Prototype notices explain that the app uses a demo catalog rather than live store inventory.
- Match labels help customers interpret whether a recommendation is a strong match or a broader fallback.
- Customer sessions remain private, while owner intelligence uses de-identified demand signals.
- Firestore rules prevent customers from changing their own role and validate accepted signal fields.
- Owner dashboard language frames signals as review inputs rather than purchase instructions.
- Human review remains required before inventory decisions.

Governance and Responsibility

The governance design separates normal operation, issue response, technical investigation, and final business authority. This separation is important because The Vinyl Concierge affects both customer trust and owner buying decisions. A store employee should be able to respond to a customer issue, but not silently rewrite core recommendation logic. A product owner should investigate and roll back technical issues, but serious incidents should remain under the authority of the business owner.

Situation	Responsible Party	Can Do	Cannot Do
Normal operation	Product owner / developer	Deploy updates, refresh catalog, monitor backend health	Ignore alerts or alter private customer data
Customer complaint	Store operator	Re-run recommendation, log complaint, respond to customer	Rewrite core logic or delete complaint history
Questionable recommendation	Store operator + product owner	Flag item, review matching logic, suppress bad record if needed	Ship logic changes without review
Demand manipulation or spam	Product owner	Down-weight or quarantine suspicious signals	Treat suspicious activity as confirmed purchase intent
Inventory mismatch	Product owner / developer	Pause inventory-sensitive recommendations and investigate	Leave corrupted recommendations running
Severe incident	Business owner	Use kill switch, roll back, or take system offline	Delay action until the next scheduled release

Early warning indicators should include recommendation complaint rate, human override rate, unusual demand spikes, repeated weak-match results, stale inventory syncs in a future production version, role or permission errors, and repeated governance interventions. The practical question is not whether the product can avoid every mistake; it is whether the organization can notice, contain, and correct mistakes before they become customer or business harm.

Business and Deployment Judgment

Who Would Pay

The realistic paying customer is the independent record store, not the individual shopper. A store would subscribe to The Vinyl Concierge to offer AI-assisted discovery to customers while receiving owner intelligence from aggregate recommendation activity. The product is most attractive for stores that want a differentiated digital experience but do not have the resources to build custom recommendation, analytics, and governance tooling themselves.

Qualitative ROI

The ROI case depends on whether the platform can improve customer engagement, repeat visits, saved-interest behavior, and inventory decisions. A simplified Week 7 break-even judgment estimated monthly operating costs around \$2,500 and suggested that a \$50 monthly price per store could be realistic if the product serves approximately 50 stores in an early pilot. This is not a full financial model, but it suggests the pilot economics are plausible if the product demonstrably improves engagement or buying decisions.

Cost / Value Area	Business Judgment
AI and cloud usage	Likely manageable with usage controls and managed infrastructure.
Support and governance	Material ongoing cost because trust, monitoring, and quality review continue after launch.
Inventory and commerce integrations	Necessary for production value but more complex than the current prototype.
Customer engagement	Potentially strong if the experience remains distinctive, useful, and transparent.
Owner decision support	Strongest differentiator, especially when future purchase and inventory validation are added.

Deployment Recommendation

The Vinyl Concierge is worth deploying as a limited pilot, not as a broad commercial release. A pilot should use one or a small number of stores, a clearly labeled catalog, explicit staff review, and careful measurement of recommendation completion, saved-interest actions, repeat visits, customer satisfaction, and owner usefulness. Broad deployment should wait until the product connects to live inventory and purchase data, because that is what would turn interest signals into stronger business intelligence.

The final business judgment is cautious but positive. The prototype has moved far enough to show a coherent product thesis: AI discovery can create operational insight when the system is designed around a feedback loop. The next investment should focus less on adding decorative features and more on proving that the loop improves real store decisions.

Final Reflection

The most important lesson from this project is that AI product value often appears around the generated output, not only inside it. The original idea was to recommend records. The stronger product became a system that uses recommendation activity to help a store understand customer interest. That shift changed the product from a themed recommendation tool into a decision-support platform.

The project also showed why responsible product judgment matters. It would have been easy to make the prototype look more complete than it was by using cart language, live inventory language, or confident owner-demand claims. The better decision was to make limits visible. The product is stronger because it says what is built, what is synthetic, what is directional, and what remains future work.

If continued after the course, I would focus on a small pilot with real store inventory, purchase validation, and staff review. I would measure whether customers complete the recommendation flow, whether saved-interest signals predict any buying behavior, whether owners find the dashboard useful, and whether the system improves decisions without replacing human judgment. That is the point at which The Vinyl Concierge would move from a promising prototype to a deployable business product.